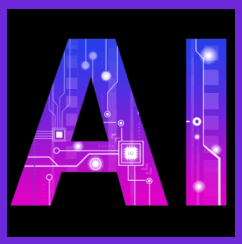
YOUTH in AI

DATA SCIENCE & AI TRAINING

Learning Hub Guide

For trainers and learners

How to use the Learning Hub, what to install for each phase, and how trainers run sessions, slides and assessment.



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Inside this guide

Part A Everyone: opening and finding your way around the Learning Hub

Part B Learners: what to install, phase by phase, and how to stay safe

Part C Trainers: before the cohort, each week, slides, assessment and troubleshooting

PART A · EVERYONE

Getting started with the Learning Hub

The Learning Hub holds the whole programme: six phases and twenty modules, each with a lesson, a practice task, a quiz and further reading. It runs in a web browser, so there is nothing to install to use it.

The six phases

Phase	You learn	Modules
1 · Foundations	What AI is, Python, databases and SQL	1.1–1.3
2 · Version Control	Git, GitHub, teamwork and your portfolio	2.1–2.3
3 · Data Science	Cleaning and analysing data, charts and dashboards, real Ghanaian case studies	3.1–3.3
4 · Machine Learning	Predictive models, the cloud, putting a model online	4.1–4.3
5 · Prompt Engineering	Getting reliable, safe results from AI models	5.1–5.4
6 · Advanced AI	Large language models, AI agents, ethics and the capstone project	6.1–6.4

Opening the hub

- Open the link your trainer sends you. The programme lead shares it from the hub's Share menu, so it only works once it has been shared with you: https://claude.ai/artifact/GArQmqKzi4abYEgpu1B6Ui
- Trainers can also send the file **data-ai-learning-hub.html**. Double-click it to open it in your browser, even without a claude.ai account.
- Use an up-to-date browser: Chrome, Edge, Firefox or Safari. The hub works on laptops, desktops, tablets and phones.
- The first time you open it you need an internet connection, because it loads its fonts and tools online.

Finding your way around

Where	What it does
Sidebar (laptop) or Modules button (phone)	Lists every phase and module. Tap a phase name to open or close it; **Collapse all** closes them all at once. A tick shows modules you have completed.
Search box	Finds modules by title, summary or key term, e.g. “pandas” or “prompt injection”.
Home page phase tabs	**All phases**, **Phase 1–6** and **Trainer guide**. Each phase card lists its modules and has the slide download button.
Module tabs	**Lesson** (the teaching content), **Practice** (a task with a hidden model answer), **Quiz** and **Resources** (further reading).
On this page bar	Jumps to any section of a lesson, including **No-code option** and **Key terms**. It stays at the top of the screen while you read.
Mark module complete	Records your progress. Scoring 75% or more on a quiz marks the module complete automatically.

Your progress

Your progress is saved **in your own browser on your own device**. It is not sent to your trainer. That means:

- Use the same device and browser each time, or your ticks will not be there.
- Private or incognito windows forget your progress when you close them.
- Clearing your browser history or site data erases your progress.

Tip. Each module has its own link. Copy the address after opening a module to come straight back to it later.

PART B · LEARNERS

What you need, and how to set it up

Almost everything runs in a browser and is free. You do not need a powerful computer, and you never need to enter a payment card.

Your device and connection

	Recommended	Also works
Device	A laptop or desktop (Windows, Mac or Chromebook)	A tablet or phone for reading lessons, quizzes and the no-code exercises
Internet	A stable connection for live sessions	Download notebooks and recordings to keep working offline
Accounts	A Google account (from Week 1) and a GitHub account (from Phase 2)	A school Google account, if your school allows Colab and Gemini

What to set up for each phase

Phase	Required	Optional or Advanced track	Age
1 · Foundations	Google account; Google Colab (browser); Teachable Machine (browser)	Python and Jupyter on your own computer	—
2 · Version Control	GitHub account; GitHub Desktop (Windows/Mac) or the GitHub website	Git command line; GitHub CLI	13+
3 · Data Science	Colab (pandas is already there); Looker Studio (browser)	Power BI Desktop (Windows only)	—
4 · Machine Learning	Colab (scikit-learn is already there); Gradio or Streamlit for demos	Google Cloud, using credits your trainer arranges	—
5 · Prompt Engineering	Gemini app (under 18) or Google AI Studio (18 and over)	Gemini API key and Python, 18 and over only	13+ / 18+
6 · Advanced AI	As Phase 5, plus GitHub for your capstone team	Gemini API for building apps, 18 and over only	13+ / 18+

Age rules. Google's Gemini API terms (effective 23 March 2026) say Google AI Studio and the Gemini API are for people aged 18 and over. If you are under 18, use the Gemini app at gemini.google.com for every Phase 5 exercise; the hub shows a no-code option for each one. GitHub accounts require you to be 13 or over.

Setup checklists

Week 1: before your first session

- ☐ Sign in to your Google account (create one at accounts.google.com if you need to).
- ☐ Open **colab.research.google.com**, create a notebook and run `print("Hello from Ghana!")`.
- ☐ Open the Learning Hub link and bookmark it. Start Module 1.1.
- ☐ Join your class in Google Classroom and test Zoom or Microsoft Teams with your trainer's link.

Before Phase 2: GitHub

- ☐ Create a free account at **github.com** with a professional username; it becomes part of your portfolio.
- ☐ Turn on two-factor authentication in your GitHub settings.
- ☐ Install **GitHub Desktop** from desktop.github.com (Windows or Mac), or plan to use the GitHub website in your browser.
- ☐ Advanced track: install Git from git-scm.com and run the one-time setup in Module 2.1.

Before Phase 5: AI tools

- ☐ Under 18: open **gemini.google.com** and check you can start a chat.
- ☐ 18 and over: sign in to **aistudio.google.com** and open a new chat prompt.
- ☐ Advanced track, 18 and over: create an API key in AI Studio only if your trainer asks you to, and keep it private.

Optional: Python on your own computer (Windows)

Colab is enough for the whole programme. If you want to work offline as well:

1. Download Python from **python.org/downloads**. When the installer opens, tick **Add python.exe to PATH**, then install.
2. Open PowerShell and install the course libraries:

```
pip install notebook pandas numpy matplotlib seaborn scikit-learn
```

3. Start Jupyter. It opens in your browser:

```
jupyter notebook
```

For the Advanced-track Gemini examples (18 and over only), also install Google's library and store your key on your computer. Close and reopen PowerShell afterwards:

```
pip install -U google-genai
setx GEMINI_API_KEY "paste-your-key-here"
```

Staying safe

- **Keep secrets secret.** Never share your passwords or API keys, never paste them into chats or slides, and never commit them to GitHub.
- **Protect people's data.** Do not put names, phone numbers or Ghana Card numbers into AI tools or GitHub. Ghana's Data Protection Act, 2012 (Act 843) applies.
- **Follow the AI-use policy.** You may use AI assistants, but **disclose** how you used them, be able to **explain** anything you submit, and **verify** facts and code.
- **Check what AI tells you.** AI can be confidently wrong, especially about numbers, names and local languages. Ask a fluent speaker to check anything in Twi, Ewe, Ga, Dagbani or Hausa.

Remember. Errors and confusion are normal. Read the last line of an error message, ask early, and help a classmate. Every expert started where you are now.

PART C · TRAINERS

Running the programme on the Learning Hub

What to prepare before a cohort, a weekly routine, how to use the slide decks, and how to assess and support learners.

Before the cohort

- ☐ **Share the hub.** Open the hub on claude.ai and share it with learners and co-trainers from the Share menu, or send them the HTML file.
- ☐ **Check ages and assign tools.** Record each learner's age band and use the age table below. Under-18 learners use the Gemini app in Phase 5.
- ☐ **Download the decks.** Use **Download Phase N slides** on each phase card, or copy them from the shared slides folder. Open each one to check it works in PowerPoint or Google Slides.
- ☐ **Test the coded AI examples.** With the programme's own key (18 and over), run **test_gemini.py**. Check the current model names at ai.google.dev/gemini-api/docs/models; Google changes them often.
- ☐ **Click through the no-code steps.** Try the AI Studio and Gemini app steps in Module 5.1 and the GitHub website steps in Module 2.1, as screens change.
- ☐ **Pass your teach-backs.** Before teaching a phase, complete its teach-back with the Technical Lead (Trainer guide tab, ToT teach-back sessions).
- ☐ **Arrange access.** Data bundles for learners who need them, cloud credits for the Advanced track, and a lab venue for in-person days.

Who can use which tool

Tool	Who can use it	Use it for
GitHub (website, Desktop, Git)	13 and over (GitHub Terms of Service)	Version control, teamwork and portfolios from Phase 2
Gemini app (gemini.google.com)	Personal Google account 13 and over; school accounts where the admin has switched Gemini on	No-code prompt practice for under-18 learners
Google AI Studio	18 and over	No-code prompt practice; trainer demos
Gemini API key (Python)	18 and over	Coded examples in Modules 5.1–5.4 and 6.1; Advanced-track capstones

Google's terms also bar Gemini API apps aimed at, or likely to be used by, under-18s. Capstone teams that build on the API need an adult key holder. Re-check the terms before each cohort.

A weekly routine

When	What to do
Before the session	Read the module's Lesson and Practice tabs, including the model answer and quiz. Check the deck's speaker notes. Prepare one local example or dataset to open with.
During the session	Short recap quiz, a brief concept explanation with the slides, then a guided lab using the Practice tab, and a reflection to close. Aim for at least 70% learner activity, averaged over the phase. Point learners without laptops, or under 18, to the no-code options.
After the session	Give feedback on submitted work within one week, as the Pedagogical Support Framework sets out, and sooner for weekly exercises. Log questions you could not answer for the weekly reflection meeting.

Using the slide decks

- Each phase has one PowerPoint deck: lessons, practice tasks, quizzes, answer keys and further reading. Download it from the phase card or from the top of any module in that phase.
- On claude.ai the browser asks you to confirm the download. When the HTML file is opened directly, the deck downloads straight away.
- Module title slides carry the learning objectives in the speaker notes.
- **Model answer** and **Quiz answer key** slides are for after learners have tried the task. Hide or skip them until then.
- Edit the decks freely in PowerPoint or Google Slides. When the hub content is updated, download the decks again so slides and hub match.

Assessment and progress

- **Quizzes:** a score of 75% or more marks a module complete in the learner's own browser.
- **Trainers cannot see learners' progress in the hub**, because it is stored only on each learner's device. Collect evidence through Google Classroom and, from Phase 2, through each learner's GitHub repositories.
- **Mini-projects and capstone:** use the rubrics in the curriculum and the Trainer guide tab. From Phase 2, check for clear commits, reviewed pull requests and no secrets in repositories.
- **Teach-back:** each phase's prompts are in the Trainer guide tab. Facilitators need 3 out of 4 or more on every criterion to teach that phase.

Troubleshooting

Problem	Fix
The hub link says the page is unavailable	It has not been shared with that person yet. Share it from the hub's Share menu, or send the HTML file.
The hub looks unstyled or blank	It needs internet the first time it opens. Reconnect, refresh, or try another browser.
A learner's progress has disappeared	They are on a different device or browser, used a private window, or cleared browser data. Progress cannot be recovered; they can re-mark modules.
"Downloads aren't available in this view"	Open the hub on claude.ai, or open the HTML file directly in a browser.
Gemini code shows 404 or "model not found"	The model name has changed. Check the models page and update the model name in the example.
Gemini code says there is no attribute 'interactions'	The library is out of date. Run pip install -U google-genai.
Git asks for a GitHub password	GitHub no longer accepts account passwords for Git. Use GitHub Desktop, or sign in with gh auth login.
An under-18 learner cannot use AI Studio	That is Google's age rule. Use the Gemini app version of the exercise and the trainer demo.

Programme contacts

Role	Name	Phone or email
Learning Design and Education Lead		
Technical Lead		
Programme coordinator		

Fill in before printing or sharing.